Thermodynamics Formula Sheet

Exam 1

Units and Conversions

SI Prefixes

femto (f)	$= 10^{-15}$	deca (da)	$= 10^{1}$
pico (p)	$= 10^{-12}$	hecto (h)	$= 10^{2}$
nano (n)	$= 10^{-9}$	kilo (k)	$= 10^{3}$
micro (μ)	$= 10^{-6}$	mega (M)	$= 10^{6}$
milli (m)	$= 10^{-3}$	giga (G)	$= 10^{9}$
centi (c)	$= 10^{-2}$	tera (T)	$= 10^{12}$
deci (d)	$= 10^{-1}$	peta (P)	$= 10^{15}$

Atmospheric Pressure

$$\begin{aligned} 1\text{ atm} &= 101.325\text{ kPa} \\ &= 1.01325\text{ bar} \\ &= 760\text{ Torr} \\ &= 760\text{ mmHg} \\ &= 29.92\text{ inHg} \\ &= 14.696\text{ psi} \\ &= 10.33\text{ m } H_2O\text{ (at } 4°\text{C)} \end{aligned}$$

Pressure Conversions

$$1\,\text{Pa} = 1\,\text{N/m}^2$$

$$1\ \text{bar} = 10^5\ \text{Pa}$$

$$1\ \text{psi} = 6.89476 \times 10^3\ \text{Pa}$$

$$1\ \text{Torr} = 133.322\ \text{Pa}$$

$$1\ \text{mmHg} = 133.322\ \text{Pa}$$

$$1\ \text{inHg} = 3.38639 \times 10^3\ \text{Pa}$$

Gauge vs Absolute Pressure

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmospheric}}$$

$$P_{\text{vacuum}} = P_{\text{atmospheric}} - P_{\text{absolute}}$$

Note: Generally, gauge pressures already account for atmospheric pressure, and thus reads zero when open to the atmosphere.

Temperature Conversions

- Absolute temperature conversions
  - $T_K = T_{\circ C} + 273.15$
  - $T_{\circ R} = T_{\circ F} + 459.67$
  - $T_{\circ F} = \frac{9}{5} T_{\circ C} + 32$
  - $T_{\circ R} = \frac{9}{5} T_K$
- Temperature difference conversions
  - $\Delta K = \Delta°C$
  - $\Delta°R = \Delta°F$
  - $\Delta°F = \frac{9}{5}\Delta°C$
  - $\Delta°R = \frac{9}{5}\Delta K$

Density and Specific Gravity

$$\rho = \frac{m}{\cancel{V}} \quad \text{(density)}, \qquad \cancel{v} = \frac{1}{\rho} = \frac{\cancel{V}}{m} \quad \text{(specific volume)}$$

Specific gravity is defined as a relative density compared to water (at 4°C) where $\rho_{water} = 1000\,\text{kg/m}^3$:

$$\text{SG} = \frac{\rho_{fluid}}{\rho_{water}} \implies \rho_{fluid} = SG \cdot \rho_{water}$$

Specific Weight

$$\gamma_s = \rho g \quad \text{(specific weight)}$$

Where hydrostatic pressure can be written as $P = \gamma_s h$

Volume

$$1\,\text{m}^3 = \underline{1000\,\text{L}} = 61024\,\text{in}^3 = 35.315\,\text{ft}^3 = \underline{264.17\,\text{gal}}$$

Slugs and lbm

$$\left[\frac{1\,\text{lbf}}{32.174\,\frac{\text{lbm}\cdot\text{ft}}{\text{s}^2}}\right], \quad \left[\frac{1\,\text{slug}}{32.174\,\text{lb},}\right], \quad \left[\frac{1\,\text{lbf}}{1\,\frac{\text{slug}\cdot\text{ft}}{\text{s}^2}}\right]$$

Energy Units

1 Btu = 778.169 ft·lbf = 1055.06 J

Work Units

1 hp = 550 ft·lbf/s = 745.7 W

Distance

1 mile = 5280 ft

Ideal Gas Equation of State

Refer to table A-4 for gas constants of common gases.

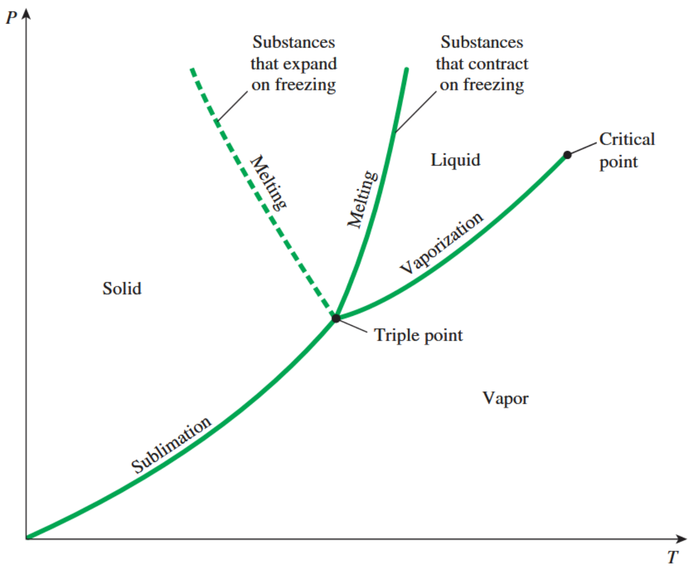
$$Pv = RT, \quad R = \frac{\bar{R}}{M} \quad \left(\text{kJ/kg}\cdot\text{K} \quad \text{or} \quad \frac{\text{Btu/lbm}\cdot\text{R}}{\text{psia}\cdot\text{ft}^3/\text{lbm}\cdot\text{R}}\right)$$

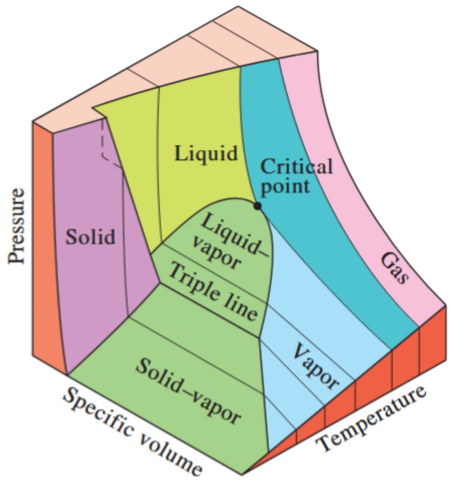
$$P\cancel{V} = mRT \quad \text{(in terms of TOTAL volume and mass)}$$

Universal gas constant: $\bar{R} = 8.314$ J/(mol·K)

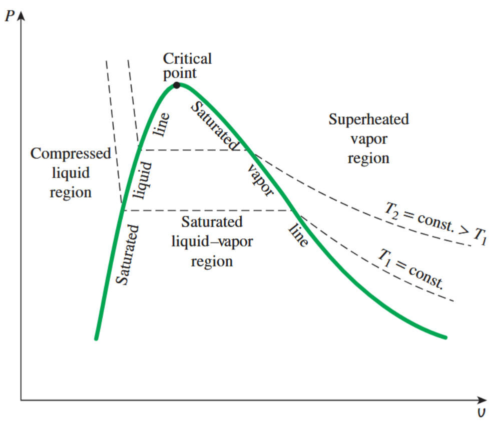
Note: Convert units for P, $\cancel{V}$, $\cancel{v}$, and T to match the units of R when using the ideal gas law. Also,

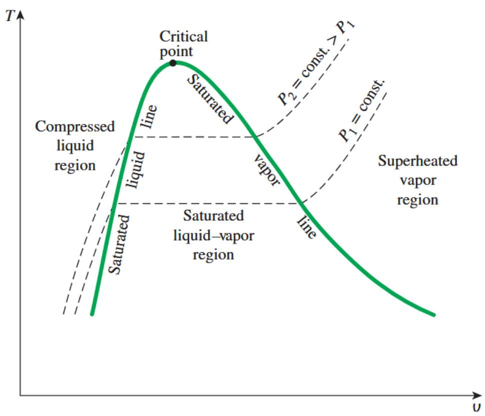
$$\left[\frac{\text{kJ}}{\text{kpa}}\right] = [\text{m}^3]$$

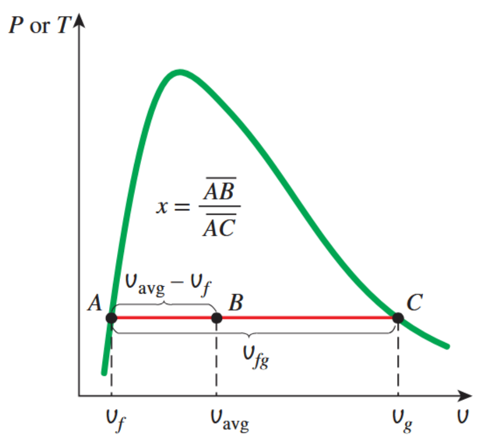




P-υ-*T* surface of a substance that expands on freezing (like water).







Pressure Relations

$P = \frac{F}{A} = \rho g h$ (hydrostatic pressure)

$P_1 = P_2 \quad \Rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2}$ (Pascal's Law)

$\Delta P = \rho g h$ (multi-fluid manometer)

(add downward, subtract upward)

Energy Definitions

$e = u + \frac{1}{2}v^2 + gh$ (specific total energy = internal + KE + PE)

$e_{\text{mech}} = \frac{P}{\rho} + \frac{1}{2}v^2 + gh$ (specific mechanical energy)

First Law (Closed System)

$\Delta U = Q - W$ (energy balance for closed system)

$\dot{E}_{in} - \dot{E}_{out} = \frac{dE_{system}}{dt}$ (rate form)

First Law (Open System)

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \underbrace{\sum \dot{m}_i \left[h_i + \frac{1}{2}V_i^2 + gz_i\right]}_{\text{Inlet}} - \underbrace{\sum \dot{m}_e \left[h_e + \frac{1}{2}V_e^2 + gz_e\right]}_{\text{Outlet}}$$

For steady state process with no heat transfer:

$\dot{W} = \dot{m}e_{\text{mech}}$

$\dot{W} = \dot{m}\frac{\Delta P}{\rho} = \dot{\cancel{V}}\Delta P$ (pump power w/ no change in KE or PE)

Mass & Flow Relations

$\dot{m} = \rho\dot{\cancel{V}} = \rho A v_{avg}$ (mass flow rate)

$\dot{E} = \dot{m}e$ (energy flow rate)

Enthalpy

$h = u + P\cancel{v}\,, \quad \cancel{v} = \frac{1}{\rho}$ (specific enthalpy)

Work

$W = \int F \cdot dx, \quad \dot{W} = F \cdot v$ (general)

$W = VI\Delta t, \quad \dot{W} = VI = RI^2 = \frac{V^2}{R}$ (electrical)

$W = 2\pi nT, \quad \dot{W} = 2\pi\dot{n}T$ (shaft)

$W = \frac{1}{2}k(x_2^2 - x_1^2)$ (spring work)

Heat Transfer

$Q = \int \dot{Q}\,dt = \dot{Q}\Delta t$ (total heat transferred)

$\dot{Q} = kA\frac{\Delta T}{\Delta x}$ (conduction)

$\dot{Q} = hA(T_s - T_f)$ (convection)

Phase Change & Saturation

$h_{fg} = h_g - h_f$ (latent heat of vaporization)

$x = \frac{m_g}{m_f + m_g}$ (quality definition)

$$\left.\begin{aligned} &y_{\text{avg}} = y_f + x(y_g - y_f) \quad \text{(relation for } \cancel{v}, h, u\text{)} \\ &x = \frac{y_{\text{avg}} - y_f}{y_g - y_f} \quad \text{(solve for quality)} \end{aligned}\right\} y_{fg} = y_g - y_f$$

Compressed Liquid Approximation

$y(T, P) \approx y_f(T)$ (use saturated liquid at same T)